AVISHKAR MAHESH PAWAR

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Professional Summary

Computer Science student with strong background in AI/ML infrastructure and high-performance computing. Built vector database implementing HNSW algorithm for semantic search on 1M+ vectors with 95%+ recall. Experience integrating OpenAI embeddings and LangChain for RAG pipelines. Proficient in Python, FastAPI, and optimizing computations using Numba JIT compilation. Published research paper on Sign Language Recognition using Machine Learning. Strong foundation in algorithms and data structures.

Education

MIT Academy of Engineering, Pune 2022 - 2026
B.Tech. - Computer Engineering — CGPA: 9.24/10

Projects

AI-Powered Vector Database

Python 3.11, FastAPI, NumPy, Numba

- Built a vector database implementing HNSW algorithm for approximate nearest neighbor search on 1,000,000+ vectors with 95%+ recall and <100ms query latency
- Optimized distance computations using Numba JIT compilation, achieving 10x speedup in cosine similarity calculations for high-dimensional vectors
- Integrated OpenAI embeddings (text-embedding-ada-002) and LangChain to power RAG pipeline for AI applications with context-aware responses

Event Streaming Platform

Java 17, Spring Boot, Netty, Apache Kafka Protocol

- Developed REST API using Spring Boot for cluster management, topic creation, and real-time monitoring with Prometheus integration
- Optimized producer batching and compression (Snappy/LZ4) reducing network bandwidth by 70%
- Containerized entire platform using Docker with multi-node deployment configuration

MiniDB — Distributed Database with ACID Transactions

C++17, Boost.Asio, Protocol Buffers, CMake

- Developed production-grade storage engine with B+ Tree indexing supporting range queries and point lookups with O(log n) complexity
- Implemented full query processing pipeline with SQL-like parser and executor supporting SELECT, INSERT, UPDATE, DELETE with JOIN operations across distributed tables
- Built MVCC (Multi-Version Concurrency Control) for snapshot isolation, enabling 1000+ concurrent transactions without read locks, with cost-based query optimization

Technical Skills

Languages: C++, Java, Python, JavaScript, TypeScript
Frameworks: Spring Boot, FastAPI, React, Node.js, Express.js
AI/ML: TensorFlow, PyTorch, OpenAI API, LangChain, NumPy, Numba
Databases: PostgreSQL, MySQL, MongoDB, Redis, Custom Database Implementations
Tools: Git, Docker, Linux, AWS, CMake, Maven, Prometheus
Distributed Systems: Kafka, gRPC, Protocol Buffers, Raft Consensus, Consistent Hashing
CS Fundamentals: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, Distributed Systems, System Design

Experience

OSMOS Jul 2025 - Present
Software Developer

- Worked on backend services using Node.js and developed applications using Python
- Contributed to product development from scratch, including designing, implementing, and testing core features

Core-Decimal Solutions Feb 2025 - Aug 2025
Software Developer

- Contributed to full-stack project using Vue.js, Express.js, and Sequelize with MVC architecture
- Developed reusable Vue.js components and integrated RESTful APIs with Axios

Achievements & Certifications

GATE 2026: Secured AIR 3124 with GATE Score 601 and 51.45/100 marks in CS — 3x rank improvement from GATE 2025 (AIR 9860)
GATE 2025: Qualified with AIR 9860, GATE Score: 453, Marks: 39.32/100
First Place in Technodium 25: Won first place in technical competition
Certifications: AWS Certified Cloud Practitioner, CCNA: Introduction to Networks, CCNA: Switching, Routing, and Wireless Essentials, CCNA: Enterprise Networking, Security, and Automation, Python Essentials 1

Extra-Curricular Activities

Outdoor Sports & Games: Active participant in outdoor sports and recreational activities